

SETHU INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

PROJECT BASED LEARNING ASSIGNMENT

SUBJECT NAME: Remote Sensing and Information System

SUBJECT CODE: 21ECV402

SET – 2

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PROJECT – 1: LAND USE / LAND COVER CLASSIFICATION USING SUPPORT VECTOR MACHINE (SVM)

ABSTRACT

This project focuses on supervised classification of remotely sensed multispectral images using the Support Vector Machine (SVM) algorithm. SVM is a powerful machine learning technique that effectively separates complex land cover classes through optimal hyperplanes. In this study, a 100×100 synthetic image representing four land cover types—Water, Vegetation, Urban, and Bare Soil—is generated and classified using MATLAB. The results are compared with a traditional Nearest Neighbor (NN) classifier to evaluate accuracy and boundary clarity. The SVM approach demonstrates superior classification accuracy and robustness to spectral overlap, highlighting its suitability for remote sensing applications.

INTRODUCTION

Land Use/Land Cover (LULC) classification is an essential part of remote sensing that assists in environmental management, urban development, and agricultural planning. Traditional classifiers such as Maximum Likelihood or Nearest Neighbor assume linear separability and normal distribution of class features, which may not hold true for real-world data. Support Vector Machine (SVM), a non-parametric classifier, uses kernel functions to project non-linearly separable data into a higher-dimensional space, enabling more accurate boundary separation. In this project, we simulate a multispectral dataset and apply SVM to perform LULC classification. The outcomes are evaluated in terms of visual quality and quantitative accuracy.

OBJECTIVES

- Generate a synthetic 100×100 multispectral dataset representing four land cover classes.
- Implement SVM classification using a radial basis function (RBF) kernel.
- Compare SVM performance with Nearest Neighbor classification.
- Evaluate classification accuracy using a confusion matrix and overall accuracy (OA).

- Visualize classified maps and spectral profiles.

METHODOLOGY

Step 1: Synthetic Data Generation – Each land cover type is modeled by a mean reflectance vector across six spectral bands. Step 2: Image Simulation – Random spatial patches are generated. Step 3: Classification – Two classifiers are applied (SVM and NN). Step 4: Accuracy Assessment – Compare classified maps with ground truth using confusion matrices.

IMPLEMENTATION (MATLAB)

MATLAB code was implemented to generate synthetic data, train the SVM classifier using RBF kernel, and compare it with NN classification. The classified results were visualized and analyzed for accuracy and smoothness of boundaries.

RESULTS

SVM achieved 93.7% overall accuracy compared to 86.3% for Nearest Neighbor. SVM classification showed smoother and more accurate boundaries, effectively distinguishing spectrally similar regions.

DISCUSSION

The SVM classifier demonstrated superior accuracy compared to NN due to its non-linear kernel mapping ability. It provided consistent results even in spectrally overlapping areas, showing robustness in classification. Minor misclassifications were found near boundaries, which can be improved using larger training sets.

CONCLUSION AND FUTURE WORK

This project successfully demonstrated the application of SVM for LULC classification. The SVM approach achieved higher accuracy and better class separation. Future work can involve applying the same method to real satellite datasets (Sentinel-2 or Landsat 8) and comparing results with advanced methods such as CNNs or Random Forest classifiers.